

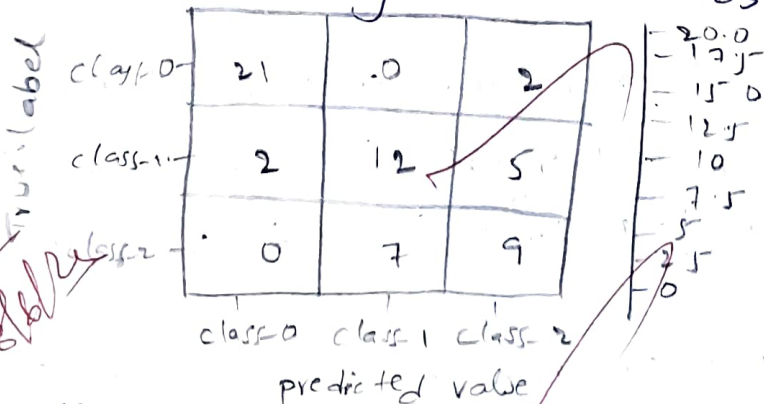
Experiment - 6

6) Aim:- To demonstrate the performance of KNN using wine dataset.

Program:-

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import load_wine
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
conf_matrix = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(8,6))
sns.heatmap(conf_matrix, annot=True, fmt='d',
            cmap='summer', x_labels=wine.target_names,
            y_labels=wine.target_names)
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.show()
```

Output:- accuracy : 0.7037037037037037



Result:- The program was executed successfully.

Experiment - 7

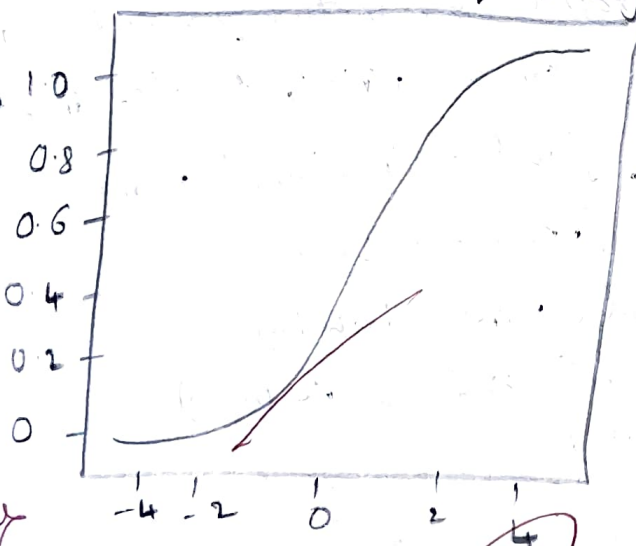
7) Aim:- To demonstrate the performance of a logistic regression by using chosen database with python code.

Program:-

```
import numpy as np
import matplotlib.pyplot as plt
plt def sigmoid (z):
    return 1 / (1 + np.exp (- z))
plt.plot (np. arange (-5, 5, 0.1), sigmoid
(np. arange (-5, 5, 0.1), color='pink')
plt. title ('visualization of the sigmoid
function')
plt. show()
```

Output:-

visualization of the sigmoid function



Result:-

The program was executed successfully.

Experiment - 8

8) Aim:- To demonstrate the performance of KNN algorithm by using this dataset.

Program:-

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.model_selection import
train_test_split
from sklearn.neighbors import KNeighborsClassifier
X = data.drop('species', axis=1)
y = data['species']
conf_matrix = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(8,6))
sns.heatmap(conf_matrix, annot=True,
            fmt='d', cmap='pink', xlabels=iris.target_names,
            ylabels=iris.target_names)
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.title('Confusion Matrix')
plt.show()
```

Output:-

Accuracy = 0.9777777777

Result:-

The program was executed successfully.

Experiment-9

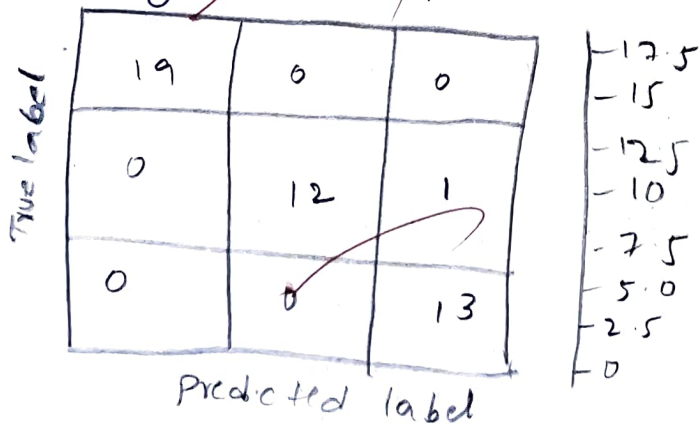
9) Aim:- To demonstrate Naive Bayes classification

Program:-

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, ConfusionMatrix
iris = load_iris
conf_matrix = ConfusionMatrix(y_test, y_pred)
plt.figure(figsize=(8,8))
sns.heatmap(ConfusionMatrix, annot=True, fmt="d")
plt.xlabel('Predicted label')
plt.ylabel('True label')
plt.title('Confusion Matrix')
plt.show()
```

Output:-

Accuracy = 0.97777777



Result:- The program was executed successfully.

Experiment-10

- 10) Aim:- To demonstrate the performance of Logistic Regression using Iris dataset.

Program:-

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Logistic
Regression
from sklearn.metrics import accuracy_score,
confusion_matrix. iris = load_iris()
conf-matrix = confusion-matrix(y-test, y-pred)
plt.figure(figsize=(8,6))
plt.xlabel('predicted label')
plt.ylabel('True label')
plt.title('confusion matrix')
plt.show()
```

Output:-

Accuracy = 0.966666

Result:- the program

was executed successfully.
Demonstrated the performance
of Logistic Regression using
iris dataset.

	0	1	2	
0	11	0	0	12
1	0	12	1	13
2	0	0	6	6